FAHAD MOLLAH

AI / Machine Learning Student | Computer Vision & Reinforcement Learning

Ingolstadt, Germany | Eligible to work up to 20h/week (student visa) | fahadmollah43@gmail.com | +49 178 6279412 | github.com/Fahad4969 | linkedin.com/in/fahad6949

PROFILE

AI Engineering master's student at THI Ingolstadt with a genuine interest in systems that can sense, decide, and act on their own. That interest has taken shape through hands-on work: training reinforcement learning agents from scratch and programming a robotic arm to sort objects in real time. Before starting the master's, spent two years in enterprise IT operations, where the focus was making sense of messy data and turning it into something people could actually act on, a habit that still shapes how the technical work gets approached today. Currently looking for a Werkstudent, internship, or thesis opportunity to keep building in AI and robotics.

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, JavaScript
- **ML / Data:** PyTorch, TensorFlow (basic), OpenCV, pandas, NumPy, scikit-learn, feature engineering, EDA, data pipelines & validation
- **AI Foundations:** Theoretical foundations of LLM / Transformer architectures
- **Autonomous Systems:** Sensor data fusion, system identification & simulation, edge computing architectures, safety-critical AI systems
- **Tools:** Git, Docker (basic), Jupyter, Linux/CLI, VS Code, VMware, Windows Server
- **Languages:** English (B2), German (A1, actively improving), Bangla (native)
- **Soft Skills:** Analytical thinking, problem-solving, teamwork, stakeholder communication

PROJECTS

Reinforcement Learning Training Pipeline for Autonomous Inspection
03/2025 – 07/2025

Python, Pygame, Q-learning, DQN

- Built a custom grid-based simulation environment to model autonomous safety-inspection tasks, then trained Q-learning and Deep Q-Network agents to operate in it, using experience replay, reward shaping, and epsilon-greedy decay to help them learn efficiently.
- Set up a reproducible training pipeline with fixed seeds and structured configs, so different agent variants could be compared fairly against each other.
- Tracked convergence, success rate, and episode length throughout training, and used those numbers to guide hyperparameter tuning and iron out instability.

Autonomous Object Sorting System (Robotics)
03/2025 – 07/2025

Arduino, Braccio, Sensors, Motion Control

- Programmed pick-and-place routines for a Braccio robotic arm, including trajectory planning and gripper control, then added a computer-vision-based color-detection module (OpenCV) so it could classify and sort objects on its own.
- Folded the arm into a multi-robot workflow and ran it under continuous operation to see how reliably it held up.

EXPERIENCE

IT Executive
07/2022 – 06/2024

Technohaven Company Limited, Dhaka

- Dug into operational data from enterprise systems to spot trends and support process-improvement decisions across teams.
- Checked data availability and quality before it went into internal reports, catching gaps before they reached stakeholders.
- Put together structured Excel reports and presentations that turned raw operational data into something decision-makers could actually use.

IT Intern
02/2022 – 04/2022

American International University, Dhaka

- Kept university IT systems and network infrastructure running, troubleshooting issues as they came up.
- Logged recurring technical issues and their fixes, building a reference that cut down repeat-issue response time.

EDUCATION

M.Eng. AI Engineering of Autonomous Systems
03/2024 – Present

Technische Hochschule Ingolstadt, Germany

- Focus on machine learning, computer vision, and robotics, with reinforcement learning (Q-learning, DQN) explored hands-on through independent projects.

B.Sc. Software Engineering
01/2018 – 06/2022

American International University, Bangladesh

- Coursework spanned software engineering, algorithms, databases, and object-oriented programming, with an early introduction to AI and expert systems.